

1. Multimodal Medical Diagnosis System

Goal: Build an AI system combining medical images, patient history, and reports.

Core Components: Vision model (ViT), Medical LLM, Fusion layer.

Requirements: Chest X-ray dataset, BioGPT/LLaMA, GPU (≥ 8 GB).

Outcomes: Diagnosis prediction, explainable AI reports, real-world healthcare application.

2. AI for Protein Design (Generative Biology)

Goal: Generate novel protein sequences with desired functions.

Core Components: Transformer/Diffusion model, Protein embeddings (ESM).

Requirements: UniProt dataset, AlphaFold outputs, high GPU compute.

Outcomes: New protein candidates, biotech/pharma applications.

3. AI That Builds AI Models (Meta-Learning System)

Goal: Automate neural architecture design.

Core Components: NAS + Reinforcement Learning.

Requirements: Search space design, compute optimization, PyTorch.

Outcomes: Optimized architectures, reduced human effort in model design.

4. Multimodal Biological Simulator (Cell-Level AI)

Goal: Simulate full cell behavior using multi-modal data.

Core Components: Graph Neural Networks, Sequence models.

Requirements: Genomic + proteomic datasets, strong compute.

Outcomes: Digital cell simulation, drug discovery acceleration.